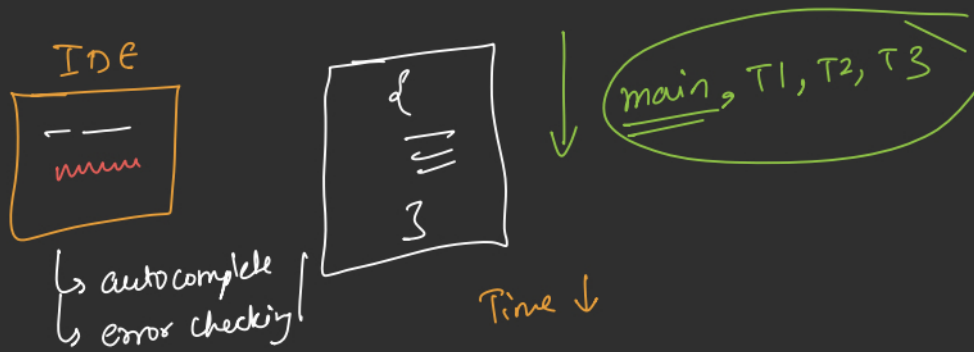
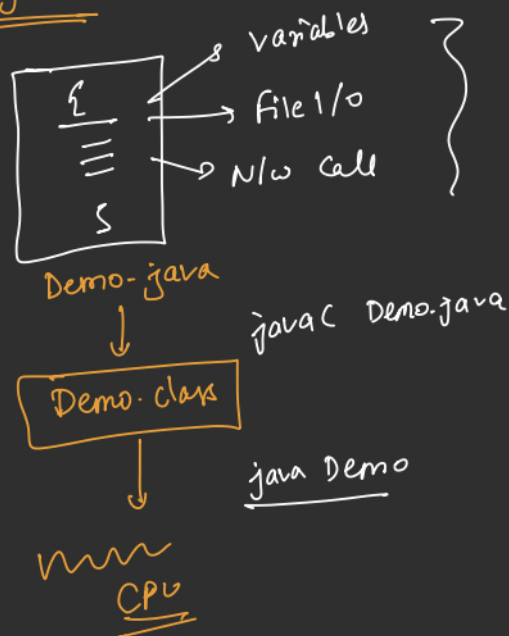


Multi Threading in Java

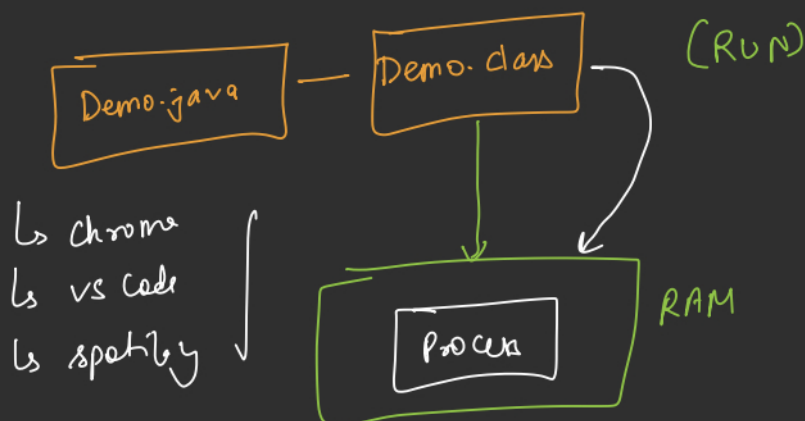


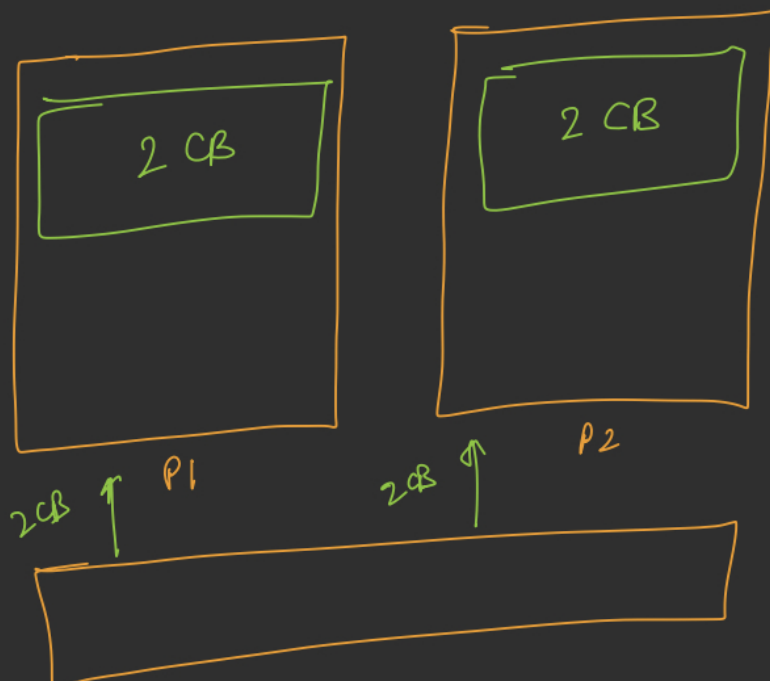
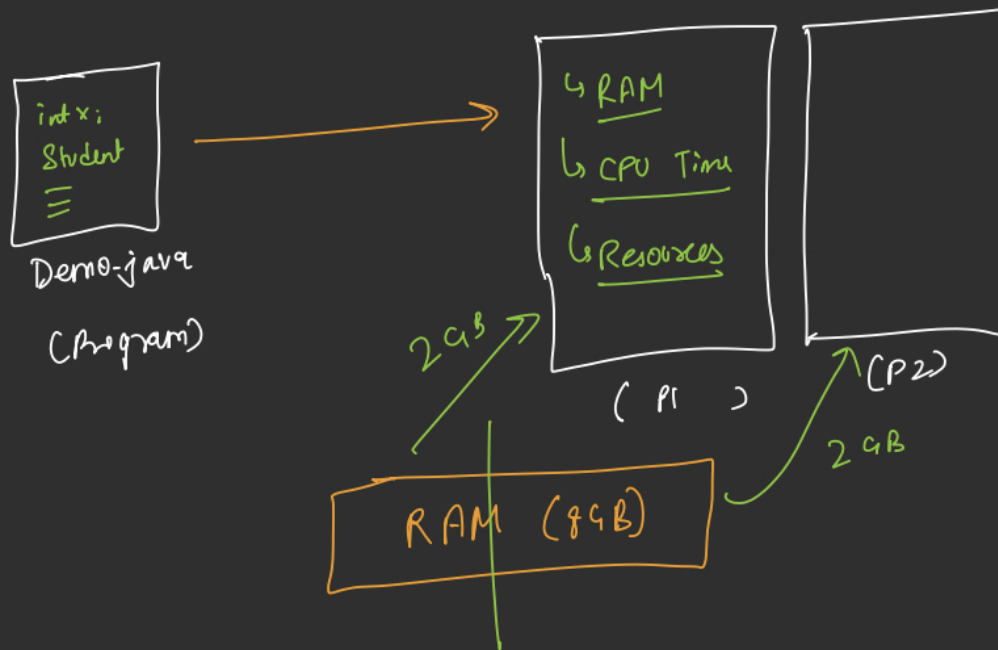
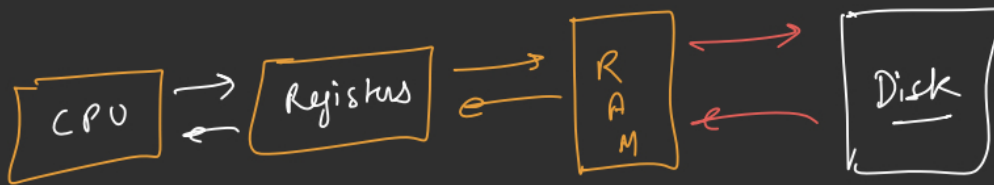
Program



Process

↳ A program that is currently being executed.





Program
↓
Process

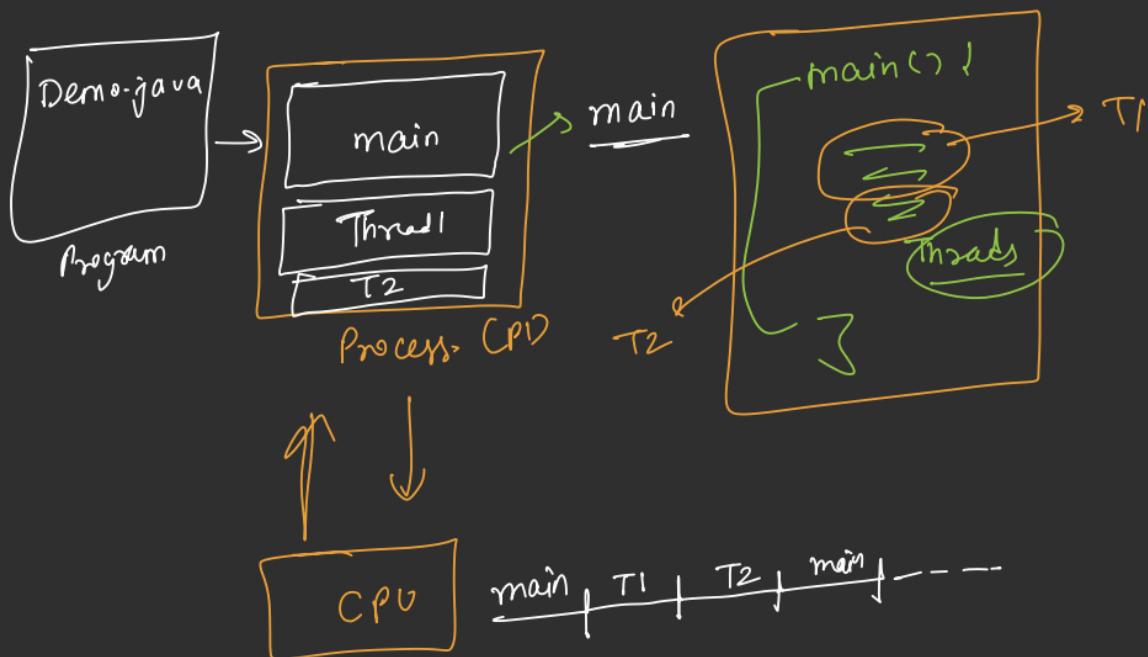
(Threads)

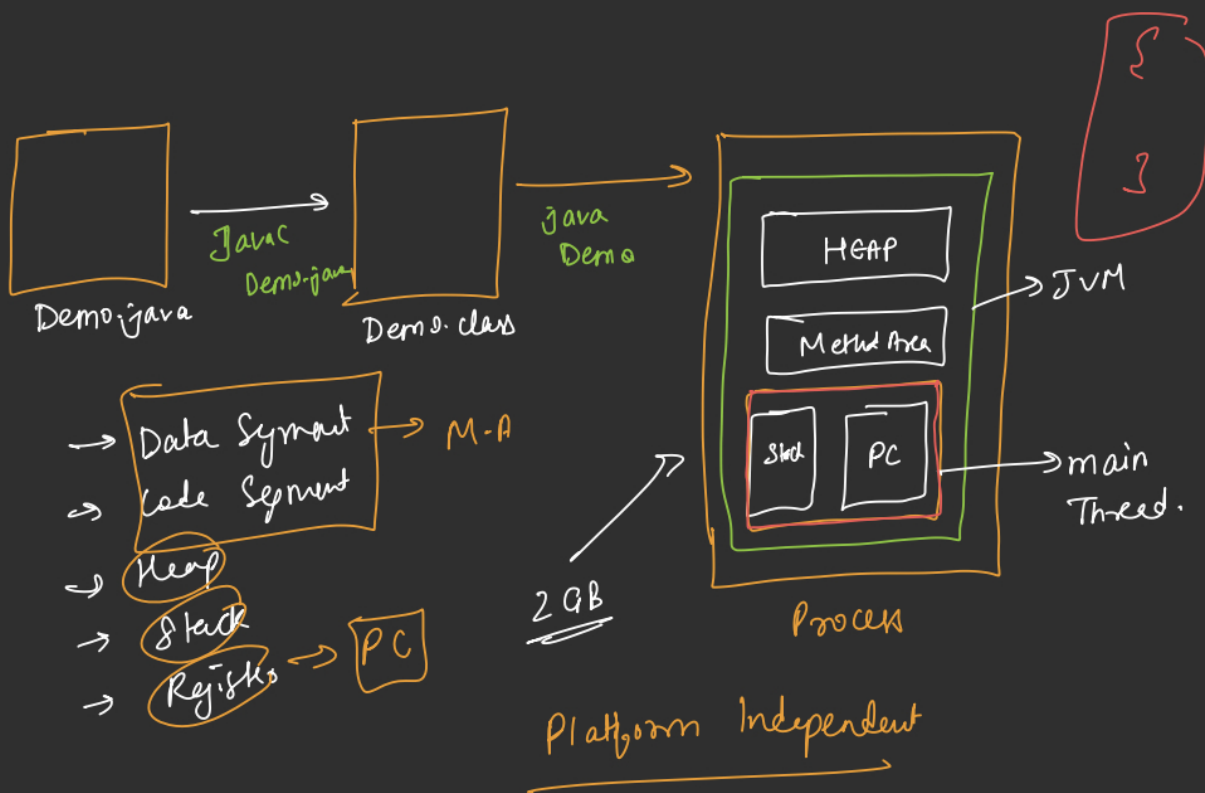
Threads

↳ Smallest sequence of instructions executed by CPU independently.

(OR)

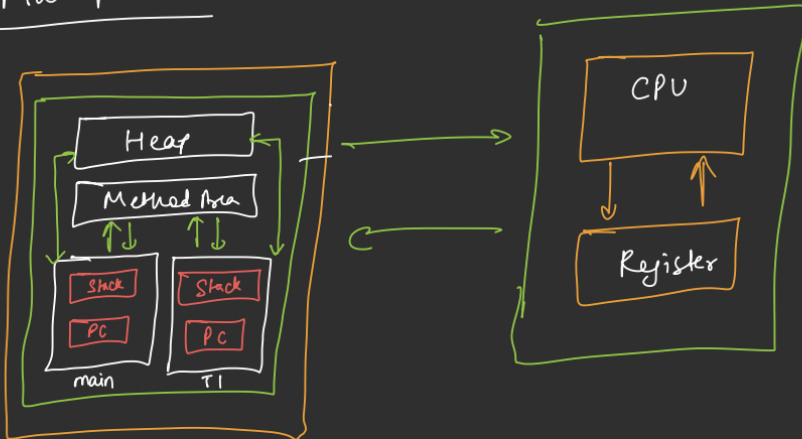
A lightweight process.



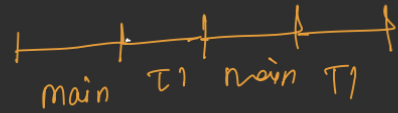


- ① JVM loads into our process.
- ② JVM uses class loader.
- ③ JVM creates a main Thread.
- ④ JVM creates separate Stack & PC space for main Thread.
- ⑤ Code starts execution from main() method.
- ⑥ main() stack frame.

Multiple Thread



Concurrent Execution



0.5 sec

IDLE

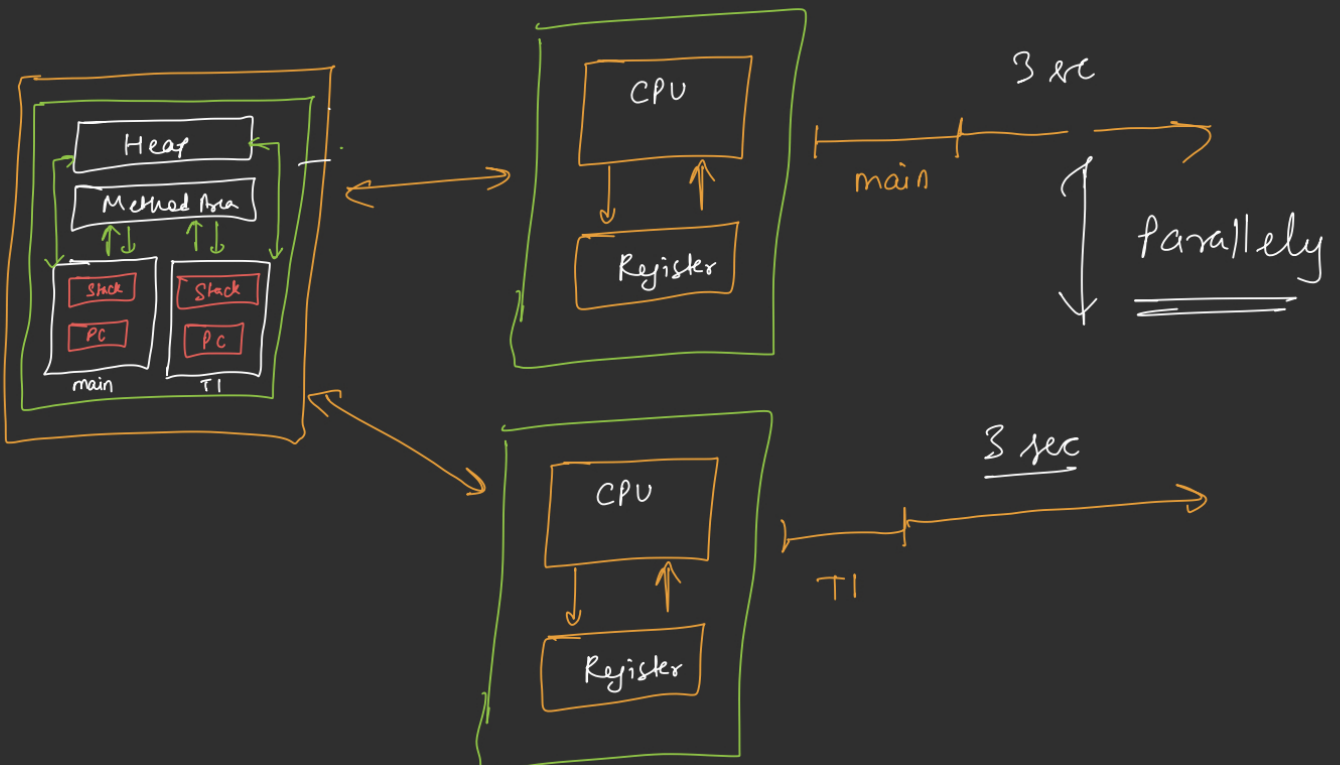
Context Switching

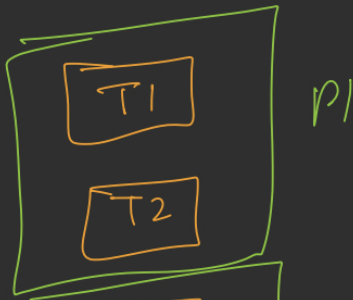
Public Class Demo {

Public static void main(String [] args) {

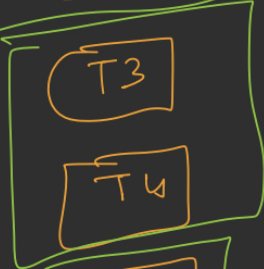


}





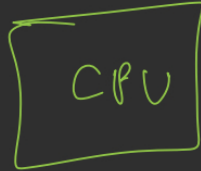
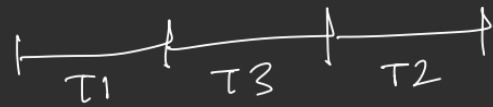
P1



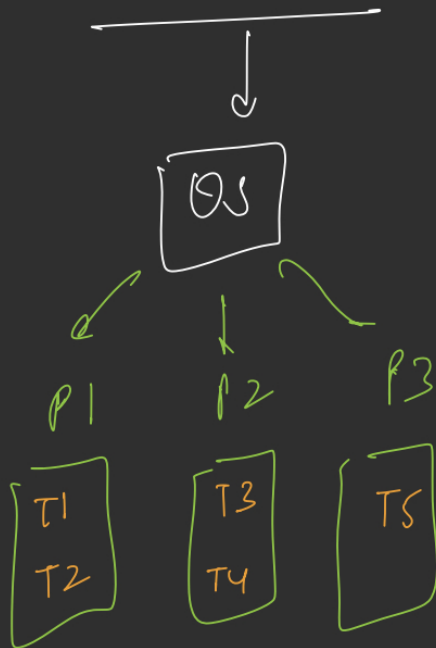
P2



P3



Multi-Tasking



Multi-Threading

